Stagecraft: IFRS 9 expected credit loss model

Model documentation · US residential mortgages (Freddie Mac loan-level data) · reporting dates 2018-12-31, 2019-12-31 · generated 19 Oct 2026

DATA NOTICE: the figures in this document were produced from the synthetic sample bundled with the repository (simulated loans written in the Freddie Mac file format, driven by real FRED macro history). They demonstrate the method end to end; they are not findings about real US mortgages. Rerunning the pipeline on the Freddie Mac sample files regenerates every number, table and chart.

1. Purpose and scope

Stagecraft estimates the impairment allowance for a portfolio of US first-lien residential mortgages under IFRS 9 (Financial Instruments, section 5.5). It allocates each loan to Stage 1, 2 or 3, estimates 12-month and lifetime probabilities of default (PD), loss given default (LGD) and exposure at default (EAD), and produces a probability-weighted expected credit loss (ECL) under base, upside and downside macroeconomic scenarios, with a severe scenario for stress testing. It also explains the movement in the allowance between two reporting dates in the format of IFRS 7.35H.

Intended use: a transparent, reproducible reference implementation for model review, teaching and interview discussion. It is not a production model: there is no governance sign-off, no management overlay and the data are a sample. Every judgment call is parameterised in `config/assumptions.yaml` and listed in section 4.

2. Data

Data	Source	Treatment
Loan-level origination and monthly performance	Freddie Mac Single-Family Loan-Level Dataset, sample files sample_2005 ... sample_2019 (about 50,000 loans per origination year). Column positions and codes follow Freddie Mac's file layout and General User Guide.	Origination: 32 fields; performance: 32 fields; pipe-delimited, no header.
Unemployment rate	FRED UNRATE	Monthly; publication lag of 1 month applied.
House prices	FRED USSTHPI (FHFA all-transactions index, US)	Quarterly; placed at mid-quarter and log-linearly interpolated to monthly; lag of 3 months applied.
Mortgage rate	FRED MORTGAGE30US	Weekly averaged to monthly; used only for the refinancing incentive.

Data checks (full report in `docs/data_report.md`): 12,000 loans across 15 vintages; 937 loans ever in default (7.8%); 355 credit-event terminations with loss data; no duplicate loan-months and no gaps in monthly histories. The sign convention of the Actual Loss field is detected from the data by rebuilding it from its components, so the LGD code works whichever sign the file uses.

2.1 Default definition and delinquency buckets

Freddie Mac reports delinquency as the number of missed monthly payments (MBA method). **Default** is the first month with status ≥ 3 (three missed payments, reported as 90-119 days), REO status 'RA', or a credit-event zero-balance code (02, 03, 09, 15: third-party sale, short sale or charge-off, REO disposition, whole-loan (non-performing) sale). This matches IFRS 9's rebuttable presumption that default does not occur later than 90 days past due (B5.5.37) and the Basel 90-day definition. The RBI's NPA test is 'overdue for more than 90 days' counted in days from the due date. At a month-end snapshot three missed payments are about 90 days overdue, so the RBI-strict reading would flag the loan a few days later, in the next monthly record (status 4). That is one switch in the configuration.

Freddie status	MBA label	Days past due at month-end	RBI class
0	Current	0 (all instalments due have been paid)	Standard
1	30-59 days	~30 (one instalment missed)	SMA-0 (1-30 days)
2	60-89 days	~60 (two instalments missed)	SMA-1 (31-60 days)
3	90-119 days	~90 (three instalments missed)	SMA-2 (61-90 days), NPA from day 91
4+ / RA	120+ days / REO	120+	NPA (sub-standard, then doubtful / loss)

A payment that is a few days late but caught up before month-end never shows in a monthly status, so SMA-0 accounts that cure within the month are invisible in this data. Exits that are neither default nor voluntary payoff (16 reperforming-loan securitisation, 96 defect repurchase) are censored.

3. Methodology

3.1 Staging

Stage	Meaning	Rule
Stage 3	Credit-impaired	In default at the reporting date, or cured less than 6 months ago (probation).
Stage 2	Significant increase in credit risk (SICR)	Any of: 30+ DPD backstop (status ≥ 1); relative test: lifetime PD now ≥ 2.0× the lifetime PD expected at origination for the same remaining term AND an increase of at least 50bp (weighted scenario PD); watchlist: modified, in forbearance / repayment / trial plan, or granted a payment deferral.
Stage 1	Performing	Everything else; 12-month ECL.

The origination PD and the current PD come from the same hazard model, so the ratio isolates the change in credit risk. The origination run uses the loan's origination score, original LTV amortised on the original schedule and indexed at long-run house-price growth, a performing status, and through-the-cycle macro inputs (no change in unemployment). The current run uses today's delinquency, modification history, HPI-indexed LTV and the scenario macro paths. Banks rarely hold the PD they actually recorded at origination for old loans; this retrospective origination PD is the usual practical substitute.

3.2 Origination scorecard

Logistic regression of default within 36 months of first payment on credit score, LTV (with a hinge at 80%), DTI, loan purpose, occupancy, channel, units, first-time buyer, number of borrowers and property type, with missing-value flags. It is trained only on loans whose whole 36-month window ended by the reporting date (7,755 loans for 2019-12-31). Its log-odds score is the borrower-risk input to the hazard model.

3.3 PD: macro-linked discrete-time hazard model (champion)

Every June and December before the reporting date, each performing loan (Current, 30 or 60 DPD) is followed monthly for up to 60 months until it defaults, exits or reaches the reporting date. Each loan-month is a row; the outcome is entry into default that month. The logistic hazard is:

```
logit h = b0 + b1·score + f(age) + g(current LTV) + Σ status-at-snapshot × horizon-bucket + modified
+ forbearance + prior default + c1·Δunemployment(12m) + c2·HPI growth(12m) + legacy-vintage flag
```

All covariates are either fixed at the snapshot or known functions of time and the macro path, so the model projects a full monthly PD term structure for every loan under any scenario without simulating future delinquency. The delinquency signal enters through status × horizon terms that decay as the horizon lengthens. Months in which a loan cannot yet reach 90 days (a current loan in its first two months) are structural zeros. Coefficients use a ridge-penalised Newton fit with step-halving. If the age slope beyond six years comes out positive it is fixed at zero, because before 2019 only crisis-era cohorts are that old. For 2019-12-31 the fit uses 2,397,374 loan-months with 3,635 default entries (unemployment coefficient 0.212 per pp; HPI growth 0.001 per %).

3.4 PD: roll-rate Markov model (challenger)

Monthly transition matrices between Current, 30 DPD, 60 DPD, Default and Prepaid, counted over the 36 months to the reporting date for nine segments (credit score band × current-LTV band). Each segment row is blended with the pooled row using a credibility weight of 200 pseudo-observations. Cumulative PD is (P^t)[s, Default]. The chain is time-homogeneous and has no macro input, so it is a sanity check rather than a scenario engine.

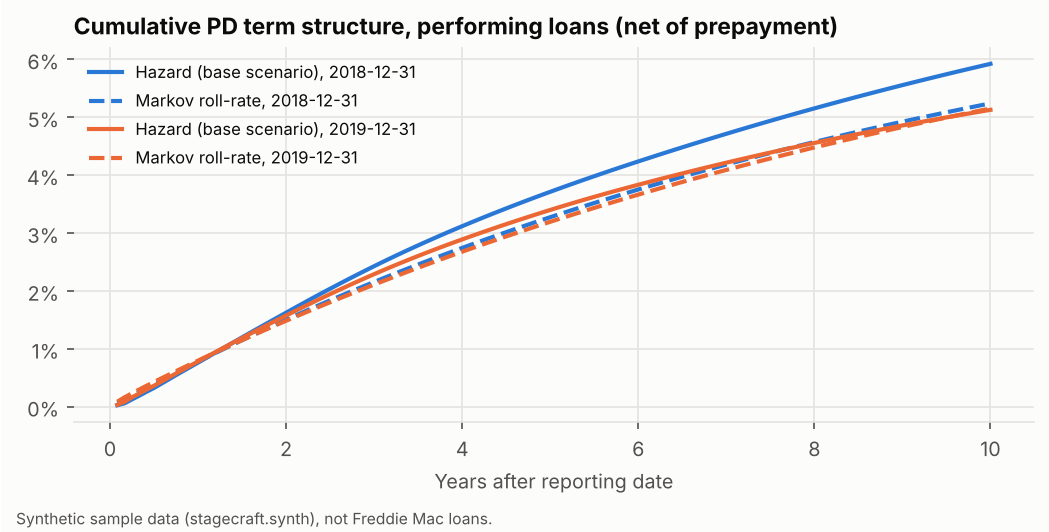


Figure 1. Average cumulative PD of performing loans: hazard model (base scenario) vs Markov chain. Both net off prepayment.

3.5 LGD

For each resolved default (credit-event exit with loss data on or before the reporting date) the economic LGD is (removal UPB + expenses + delinquent interest – recoveries discounted at the note rate over the workout months) ÷ UPB at default, where recoveries are net sale proceeds plus MI and non-MI recoveries. A fractional-logit severity model on current LTV at default (original LTV indexed by HPI), MI coverage and investor occupancy supplies the macro link: scenario house prices move projected LTV, and so LGD. Effective LGD = P(liquidation | default) × severity, floored at 5%. P(liquidation) = 46.1% for 2019-12-31, estimated from defaults followed for 24 months. Downturn LGD = mean severity of 2008-2011 defaults by LTV band. Stage 3 LGD moves the current LTV by the scenario's house-price change over a 18-month workout.

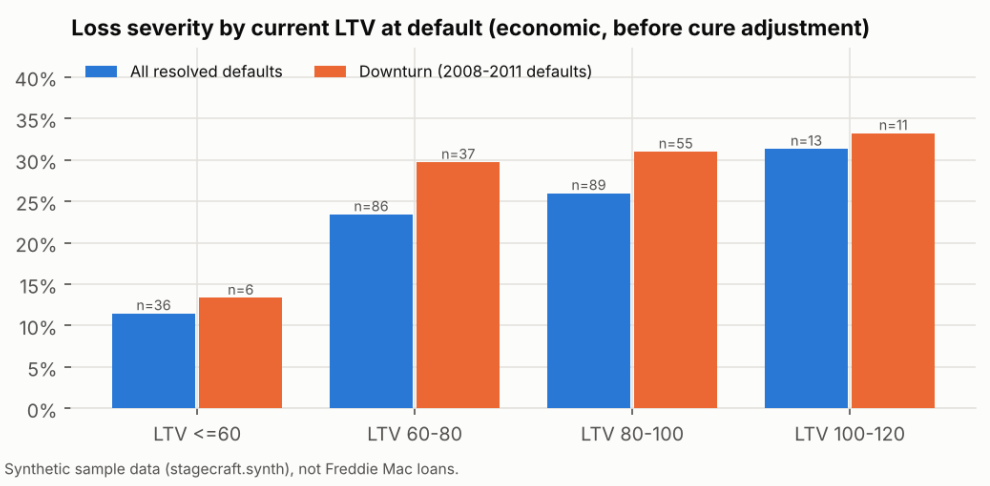


Figure 2. Loss severity by current LTV at default, all resolved defaults vs 2008-2011 defaults (model as at 2018-12-31).

3.6 EAD and prepayment

EAD in month t is the contractual balance at the start of t, from the current interest-bearing UPB, note rate and remaining term, plus any non-interest-bearing (deferred) UPB. Prepayment is a single monthly rate (SMM) per loan, estimated over the previous 36 months as the share of Current loans that pay off in full the next month, by refinancing-incentive band (note rate minus the market 30-year rate). It is applied once, as a competing exit in the survival curve.

Incentive band	loan_months	prepayments	smm	cpr
+0.5 to +1.5pp	25,178	350	1.143%	12.9%
-0.5 to +0.5pp	91,323	587	0.651%	7.5%
<-0.5pp	47,865	189	0.466%	5.4%
>+1.5pp	907	38	0.914%	10.4%

3.7 ECL

$$ECL = \Sigma_t \, S_{t-1} \cdot h_t \times LGD_t \times EAD_t \times (1 + r/12)^{-t}, \quad S_t = S_{t-1}(1 - h_t)(1 - SMM)$$

$S_{t-1} \cdot h_t$ is the marginal PD: the probability of surviving, neither defaulted nor prepaid, to month t and then defaulting in it. The marginal PDs sum to a cumulative PD of at most 1. Stage 1 sums the first 12 months; Stage 2 sums to the remaining contractual term; Stage 3 is LGD × gross carrying amount with PD = 1. Discounting uses the current note rate as the effective interest rate proxy. The probability-weighted ECL is $\Sigma \, w_s \, ECL_s$ over the weighted scenarios.

3.8 Forward-looking scenarios

Five-year paths for unemployment and house prices start from the data published at the reporting date. After year 5 they glide over 2 years to long-run values (unemployment 4.5%, HPI growth 3.0% a year). Weights: base 60%, upside 20%, downside 20%. Severe (unemployment to 10%, house prices down about 20%) is used for stress testing only.

Scenario	Unemployment, year-end Y1-Y5 (%), 2019-12-31	HPI growth Y1-Y5 (%)	Weight
base	3.6, 3.8, 4.1, 4.3, 4.5	+4.0, +3.5, +3.0, +3.0, +3.0	60%
upside	3.3, 3.3, 3.5, 3.8, 4.1	+6.0, +5.0, +4.0, +3.5, +3.0	20%
downside	5.5, 7.0, 6.5, 5.8, 5.2	-4.0, -3.0, +1.0, +2.5, +3.0	20%
severe	8.0, 10.0, 9.5, 8.5, 7.5	-12.0, -10.0, -2.0, +2.0, +3.0	stress only

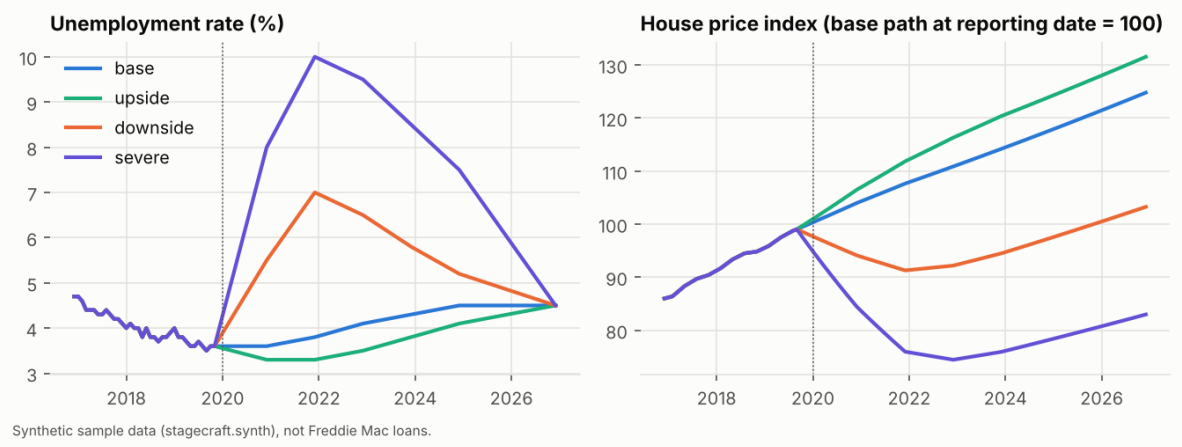


Figure 3. Scenario paths from 2019-12-31. The vertical line marks the reporting date.

4. Assumptions owned by the model owner [RISK TO REVIEW]

Each item is a judgment, not an estimate. The draft value is shown with the rationale; all live in `config/assumptions.yaml`.

Assumption	Draft value	Rationale
Default definition	status ≥ 3, RA, codes 02/03/09/15	US D90 convention; IFRS 9 90-day presumption; status 4 is the RBI-strict alternative.
Whole-loan sale (15) is a credit event	yes	These are non-performing loan sales and Freddie computes a loss on them.
30+ DPD backstop	status ≥ 1	IFRS 9 5.5.11 presumption; RBI SICR presumption at >30 DPD.
Relative SICR multiple	2.0×	Common industry setting for low-PD retail; see the cliff-effect sensitivity.
Absolute SICR floor	50bp	Stops trivially small PDs that double from triggering Stage 2.
SICR PD basis	weighted	Weighted lifetime PD reflects all scenarios in the staging decision.
Watchlist	modification, forbearance, deferral	Forbearance is a concession; IFRS 9 B5.5.17 and RBI credit-impaired indicators.
Stage 3 probation	6 months	Avoids flip-flopping between Stage 3 and 1/2 after a cure.
LGD floor	5%	Foreclosure costs and time value never go to zero.
Cure adjustment to LGD	on	About half of 90+ defaults cure; ignoring that double-counts loss.
Stage 3 workout period	18 months	Typical US foreclosure timeline; house prices move meanwhile.
Hazard seasoning constraint	no rise after age 72m	Late-age slope is confounded with crisis cohorts in pre-2019 data.
Prepayment	flat CPR by incentive band	Simple, transparent; scenario-independent.
Scenario paths and weights	base 60%, upside 20%, downside 20%	Downside as likely as upside; base most likely.
Management overlay	none	Not modelled; see Limitations.

5. Results

ECL by stage at 2018-12-31

Stage	Loans	Gross carrying amount	Share	Weighted ECL	Coverage	Base ECL	Mean 12m PD	Mean lifetime PD
Stage 1	4,776	$1.08bn	97.22%	$455,600	0.04%	$420,695	0.65%	8.28%
Stage 2	97	$18.65m	1.67%	$333,293	1.79%	$323,206	8.15%	40.18%
Stage 3	46	$12.33m	1.11%	$747,161	6.06%	$714,698	0.00%	0.00%
Total	4,919	$1.11bn	100.00%	$1.54m	0.14%	$1.46m	0.79%	8.83%

ECL by stage at 2019-12-31

Stage	Loans	Gross carrying amount	Share	Weighted ECL	Coverage	Base ECL	Mean 12m PD	Mean lifetime PD
Stage 1	5,031	$1.13bn	97.59%	$467,819	0.04%	$431,732	0.63%	6.42%

Stage	Loans	Gross carrying amount	Share	Weighted ECL	Coverage	Base ECL	Mean 12m PD	Mean lifetime PD
Stage 2	94	$17.34m	1.50%	$278,354	1.61%	$270,599	10.58%	36.79%
Stage 3	44	$10.43m	0.90%	$625,304	5.99%	$601,986	0.00%	0.00%
Total	5,169	$1.15bn	100.00%	$1.37m	0.12%	$1.30m	0.81%	6.92%

Weighted ECL is +5.1% above base ECL at 2019-12-31 (+5.3% at 2018-12-31). This is the non-linearity IFRS 9 B5.5.42 asks for: PD (logistic in unemployment and LTV) and LGD (rising with LTV, floored at the bottom) are convex in the macro drivers. The downside adds more loss than the upside removes, so the probability-weighted average exceeds the loss in the average (base) scenario. Under the severe stress the allowance would be $2.56m (1.9× the weighted figure).

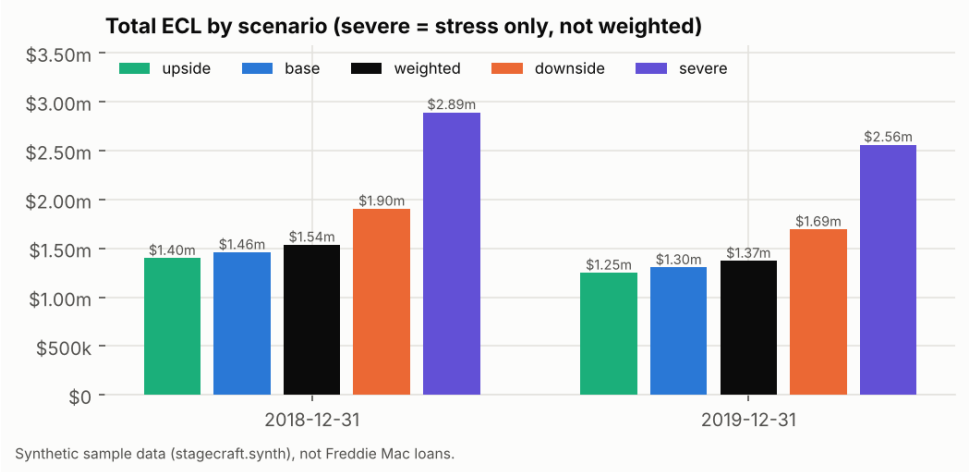


Figure 4. ECL by scenario.

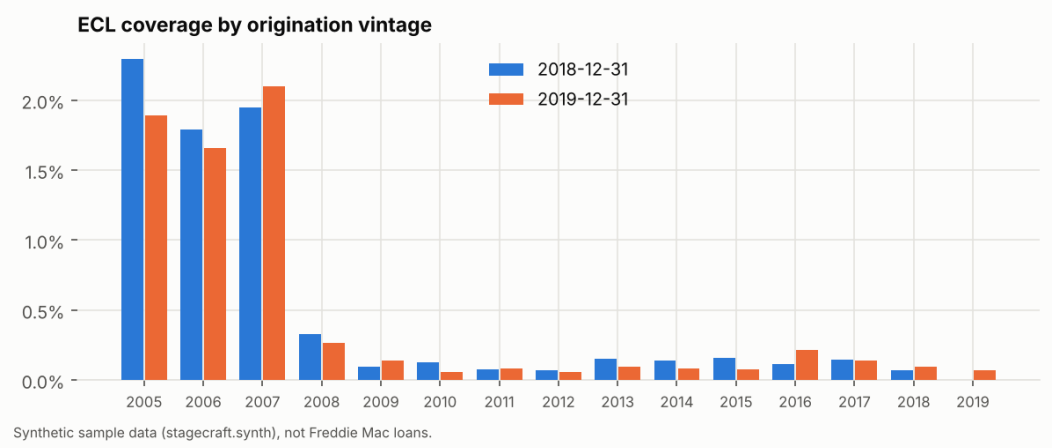


Figure 5. Coverage by origination vintage. Pre-2009 vintages carry the highest coverage (weaker underwriting, more prior delinquency).

Provision movement 2018-12-31 to 2019-12-31

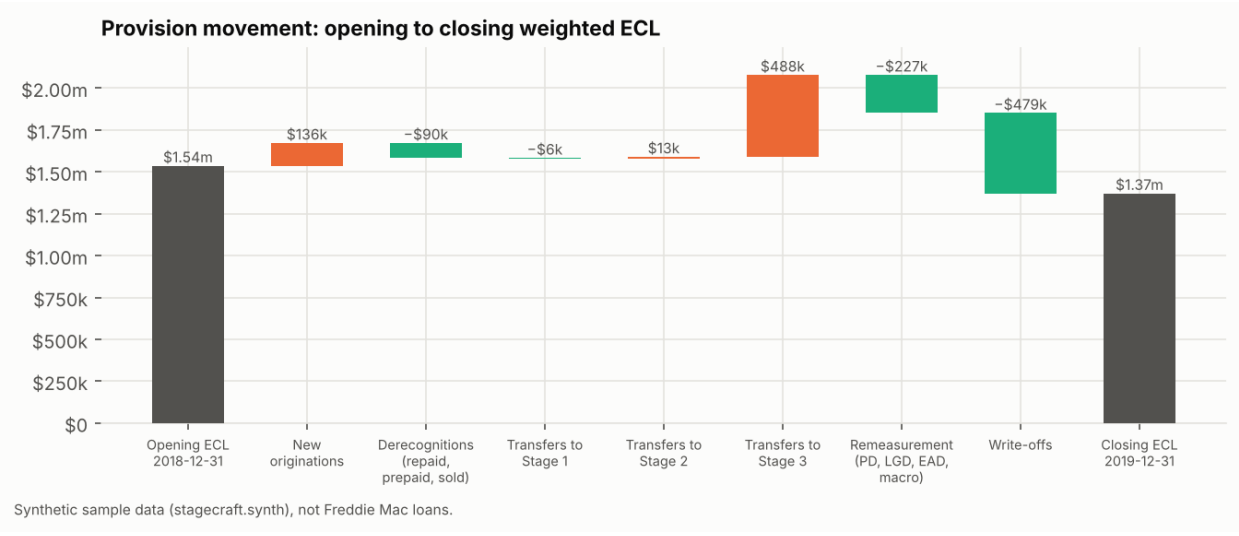


Figure 6. Provision movement waterfall (probability-weighted ECL).

Movement	Amount
Opening ECL 2018-12-31	$1.54m
New originations	$136,115
Derecognitions (repaid, prepaid, sold)	$-89,654
Transfers to Stage 1	$-6,036
Transfers to Stage 2	$13,272
Transfers to Stage 3	$488,368
Remeasurement (PD, LGD, EAD, macro)	$-227,442
Write-offs	$-479,199
Closing ECL 2019-12-31	$1.37m

Attribution is loan by loan. Loans that leave the book release their opening allowance as a derecognition or a write-off. For continuing loans, the transfer effect is the opening-date ECL recomputed under the closing stage minus the opening ECL, and remeasurement is the remainder (new PDs, LGDs, macro paths, re-estimated models, amortisation). Closing = opening + movements holds exactly, and a test enforces it. The by-stage version (IFRS 7.35H layout) is in outputs/provision_waterfall_by_stage.csv.

6. Validation (out-of-time)

6.1 PD backtest and discrimination

Predicted 12-month PD (probability-weighted) for Stage 1 and 2 loans at each reporting date, compared with default entries in the following 12 months. Data after the reporting date is used here and nowhere else. The traffic light is green if observed defaults fall inside the 95% binomial band around the predicted mean, amber inside 99.9%, red outside.

Date	Loans	Defaults	Observed	Predicted (hazard)	Predicted (Markov)	Light	AUC hazard	AUC Markov	Scorecard AUC in/OOT
2018-12-31	4,873	38	0.78%	0.80%	0.78%	green	0.793	0.750	0.786 / 0.612
2019-12-31	5,125	218	4.25%	0.82%	0.80%	red	0.599	0.585	0.780 / 0.572

From 2018-12-31 the model is well calibrated in aggregate and by segment (7 of 7 segments green). From 2019-12-31 the realised rate is several times the prediction: the 2020 COVID shock (unemployment from 3.5% to 14.8% in two months) and forbearance programmes that reported paused payments as delinquency. None of the weighted scenarios anticipated a shock of that size or speed, and the 90+ DPD definition counted forborne loans as defaulted. In practice banks handled this with new scenarios and post-model overlays during 2020. Discrimination also drops, because COVID distress was less related to the origination risk factors. The scorecard's out-of-time AUC is measured on the vintages whose 36-month window ran past the reporting date, so it is affected by the same shock.

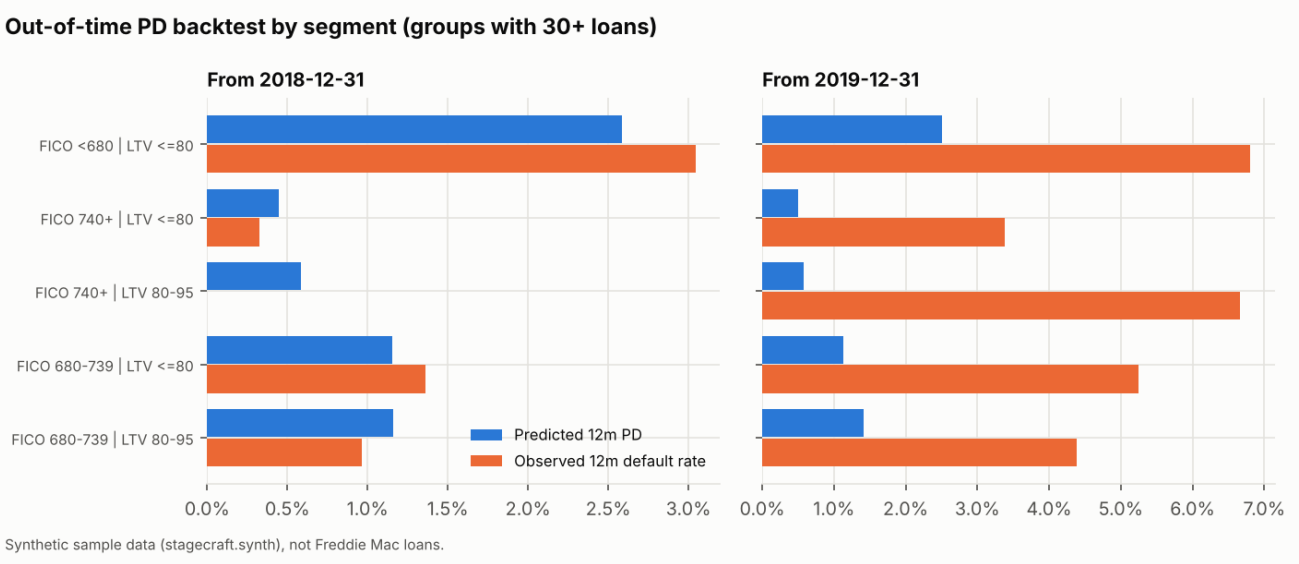


Figure 7. Backtest by segment.

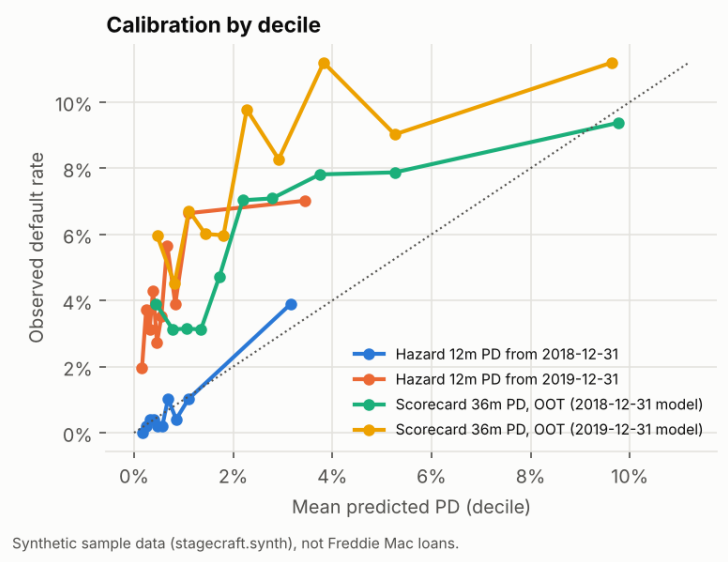
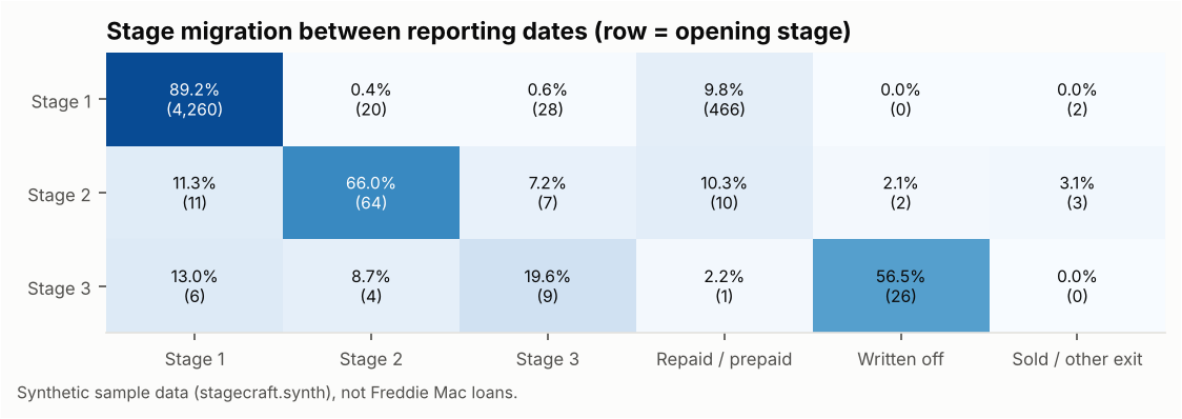


Figure 8. Calibration by decile of predicted PD.

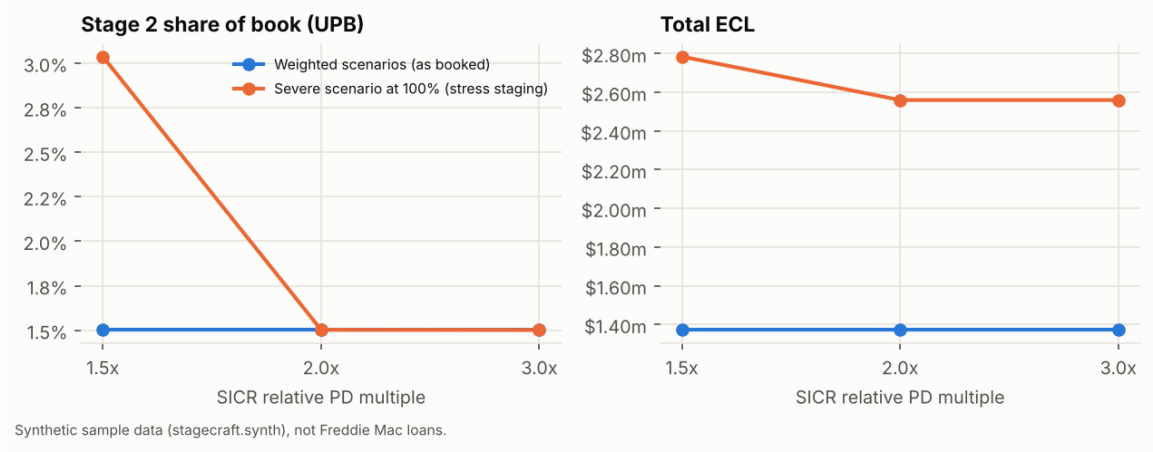
6.2 Stage migration



6.3 Sensitivities and the cliff effect

Test	Setting	ECL	vs model	Stage 2 (loans)	Stage 2 (UPB)
Model as configured	SICR 2.0x, 60/20/20	$1.37m	+0.0%	1.82%	1.50%
SICR relative threshold	1.5x	$1.37m	+0.0%	1.82%	1.50%
SICR relative threshold	2.0x	$1.37m	+0.0%	1.82%	1.50%
SICR relative threshold	3.0x	$1.37m	+0.0%	1.82%	1.50%
SICR threshold under stress staging	1.5x	$2.78m	+103.0%	3.37%	3.04%
SICR threshold under stress staging	2.0x	$2.56m	+86.6%	1.84%	1.50%
SICR threshold under stress staging	3.0x	$2.56m	+86.6%	1.82%	1.50%
Scenario weights	downside heavy (40/10/50)	$1.49m	+8.9%	1.82%	1.50%
Scenario weights	base heavy (80/10/10)	$1.34m	-2.4%	1.82%	1.50%
Scenario weights	equal (34/33/33)	$1.42m	+3.2%	1.82%	1.50%
LGD	downturn (2008-2011) LGD	$2.13m	+55.7%	1.82%	1.50%
Single scenario (100%)	base	$1.30m	-4.9%	1.82%	1.50%
Single scenario (100%)	upside	$1.25m	-8.9%	1.82%	1.50%
Single scenario (100%)	downside	$1.69m	+23.5%	1.82%	1.50%
Single scenario (100%)	severe	$2.56m	+86.6%	1.84%	1.50%

With benign base-case conditions and house prices above origination levels, almost no Current loan's lifetime PD has doubled since origination. Stage 2 is driven by the 30-DPD backstop and the watchlist, so moving the relative threshold between 1.5× and 3× barely changes the allowance (the cliff is latent). Under stress staging (severe-scenario PDs), a 1.5× threshold moves materially more loans into Stage 2. Using downturn LGD is the largest single sensitivity.



7. Limitations

- **US mortgages vs other portfolios.** Secured, amortising, low-PD loans whose losses hinge on house prices. Unsecured retail, credit cards (revolving EAD, behavioural life), SME and corporate books need different PD drivers, CCFs and LGD models.
- **Sample and synthetic data.** This run uses a simulated sample; Freddie Mac sample files are a 50,000-loan-per-year random sample of conforming loans acquired by one GSE, so they exclude jumbo, subprime and portfolio loans, and low-default segments have thin data.
- **Note rate as EIR proxy.** The effective interest rate should include fees and costs amortised over the expected life; they are not in the data. The RBI Directions allow the contractual rate on transition for the same reason.
- **Simplified prepayment.** One flat SMM per incentive band, with no burnout, seasoning ramp, turnover seasonality or scenario dependence. In reality a rate shock lowers prepayment and lengthens exposure.
- **Collateral.** Property values are indexed with the national FHFA HPI only, with no state or MSA indices, no property-level revaluation, no foreclosure-timeline differences between judicial and non-judicial states, and MI is reflected only through the severity regression.
- **No management overlay.** An overlay (post-model adjustment) is a documented, governed top-up or release of the allowance for risks the model cannot see: COVID forbearance in 2020, a new regulation, a data gap or a known model weakness. It is quantified separately, approved by a committee and reversed when the model catches up. None is booked here; the config has a placeholder (overlay.enabled).
- **Retrospective origination PD.** The origination PD is re-estimated with today's model rather than taken from the bank's records at origination.
- **Macro vintages.** FRED series are the latest revisions, not the data as first published (ALFRED). Only publication lags are respected.
- **Default definition and COVID.** Forbearance in 2020-21 inflates 90+ DPD defaults that later cured. The probation rule and the cure-adjusted LGD dampen the effect but do not remove it.
- **Model risk.** The snapshot-horizon panel has overlapping observations, so standard errors are optimistic; the ridge penalty and seasoning constraint are expert choices.

8. Relevance to India

Where the RBI stands (as of this document). On 27 April 2026 the RBI issued the final Reserve Bank of India (Commercial Banks - Asset Classification, Provisioning and Income Recognition) Directions, 2026, replacing the incurred-loss IRACP norms with an IFRS 9-style expected credit loss framework **from 1 April 2027** [1, 2]. The draft was released on 7 October 2025 [3], after a January 2023 discussion paper [4]. They apply to commercial banks other than small finance, payments and local area banks [2, para 4]. Separate directions for All India Financial Institutions will follow [1]. Cooperative banks such as UCBs are not covered by these Directions and remain on the existing norms.

- **Three stages** as in IFRS 9 (12-month ECL in Stage 1, lifetime in Stages 2 and 3) [2, para 25], with a rebuttable presumption of SICR above **30 days past due** [2, paras 33-36], the same backstop Stagecraft uses.
- **NPA stays as the credit-impaired anchor:** a term loan is NPA when overdue for more than 90 days [2, para 8]. The existing SMA-0/1/2 early-warning buckets (1-30, 31-60, 61-90 days overdue) [5] map onto Stagecraft's

Current/30/60/90 buckets as in section 2.1. SMA-1 and SMA-2 accounts are Stage 2 by the backstop.

- **Prudential floors** [2, para 82]: for housing loans to individuals, at least 0.25% (Stage 1) and 1.5% (Stage 2), and for Stage 3 10% in the first year rising to 100% after four years (higher on any unsecured portion). Stagecraft's modelled Stage 1 coverage of 0.04% would sit below the 0.25% floor, so an Indian bank would hold the floor. Modelled ECL is a minimum, not the answer.
- **Multiple probability-weighted scenarios** with documented expert judgement [2, paras 47-49, 69-72], as in section 3.8. Management overlays count as general provisions for Tier 2 capital [2, para 75].
- **EIR:** contractual rate allowed for the opening ECL, with all loans moving to EIR by 31 March 2030 [2, paras 50-54]. This is the same compromise as Stagecraft's note-rate proxy.
- **Transition:** the excess of ECL over existing provisions goes to opening retained earnings, with an optional CET1 add-back phased out by 31 March 2031 (4/5, 3/5, 2/5, 1/5) [2, paras 107-108].

Practical translation. An Indian bank's monthly SMA/NPA reporting already holds the delinquency history needed for a Markov roll-rate model. A hazard model needs the same panel plus macro links (GDP, repo rate, property prices such as the RBI's HPI), and the RBI's floors then sit on top of the modelled numbers.

[1] RBI Press Release 2026-2027/150, 27 Apr 2026, https://www.rbi.org.in/Scripts/BS_PressReleaseDisplay.aspx?prid=62628 · [2] RBI (Commercial Banks - Asset Classification, Provisioning and Income Recognition) Directions, 2026, https://www.rbi.org.in/Scripts/BS_ViewMasDirections.aspx?id=13400 · [3] RBI press release, 7 Oct 2025, https://rbi.org.in/Scripts/BS_PressReleaseDisplay.aspx?prid=61377 · [4] RBI Discussion Paper on Introduction of Expected Credit Loss Framework for Provisioning by Banks, Jan 2023 · [5] RBI circular DOR.STR.REC.68/21.04.048/2021-22, 12 Nov 2021 (IRACP clarifications, SMA classes). Accessed Oct 2026.

Appendix A. Hazard model coefficients (2019-12-31)

term	coef	std_err	z
const	-5.456	0.444	-12.3
score	0.746	0.017	44.0
age_0_12	1.031	0.433	2.4
age_12_36	0.379	0.084	4.5
age_36_72	-0.546	0.069	-7.9
age_72_180	-0.029	0.184	-0.2
cltv	-0.373	0.179	-2.1
cltv_over80	1.729	0.366	4.7
cltv_over100	1.097	0.820	1.3
modified	0.365	0.137	2.7
assisted	0.000		
prior_default	0.371	0.126	2.9
dur12	0.212	0.016	13.4
hpa12	0.001	0.008	0.1
legacy_vintage	0.562	0.068	8.2
s0_h0	-0.690	0.133	-5.2
s0_h1	-0.137	0.067	-2.0
s0_h2	-0.036	0.053	-0.7
s0_h3	-0.011	0.044	-0.2
s1_h0	4.227	0.108	39.2
s1_h1	2.010	0.213	9.5
s1_h2	0.873	0.273	3.2
s1_h3	0.286	0.296	1.0
s1_h4	0.365	0.265	1.4
s2_h0	6.227	0.148	42.2
s2_h1	2.407	0.545	4.4
s2_h2	-0.271	0.887	-0.3
s2_h3	0.821	0.679	1.2
s2_h4	0.185	0.744	0.2

sJ_hB = status J at the snapshot (0 Current, 1 = 30 DPD, 2 = 60 DPD) × horizon bucket B (≤3, 4-6, 7-12, 13-24, >24 months). Zero coefficient with no standard error = term dropped (no variation, or the seasoning constraint).

Appendix B. Reproducing this document

`pip install -e . && python -m stagecraft.pipeline` rebuilds the panel, models, outputs/ and this PDF from data/raw/ (falling back to the bundled sample). Download the Freddie Mac files sample_2005.zip ... sample_2019.zip from the Single-Family Loan-Level Dataset page (free registration) into data/raw/freddie/. Set hazard.nonevent_sample_frac to about 0.2 for the full sample to keep memory modest.